

PostgreSQL Free Version Install and Local Database Setup

Windows-focused guide for installing PostgreSQL, confirming it is free, and configuring this Spring Boot project to use a local PostgreSQL database.

Free License And Current Stable Version

PostgreSQL is free and open source. It is released under the PostgreSQL License, a liberal open source license similar to BSD or MIT.

As of June 17, 2026, the current stable major version is **PostgreSQL 18**, and the current minor release is **18.4**. PostgreSQL 19 Beta 1 exists, but beta releases are not recommended for normal local application development.

Official license page: <https://www.postgresql.org/about/licence/>

Official versioning page: <https://www.postgresql.org/support/versioning/>

Download And Install On Windows

1. Go to <https://www.postgresql.org/download/windows/>.
2. Choose Download the installer. PostgreSQL.org sends Windows users to the EDB interactive installer.
3. On the EDB download page, select PostgreSQL 18.4 for Windows x86-64.
4. Download and run the installer as Administrator.
5. Choose an install directory, normally C:\Program Files\PostgreSQL\18.
6. Select components: PostgreSQL Server, pgAdmin 4, Command Line Tools, and Stack Builder. Stack Builder is optional.
7. Choose a data directory, normally C:\Program Files\PostgreSQL\18\data.
8. Set and save the password for the built-in postgres superuser.
9. Use port 5432 unless another PostgreSQL service is already using it.
10. Use the Default Locale unless your project requires a specific locale.
11. Finish installation. You can skip Stack Builder unless you need extra tools.

Verify The Install

Open a new PowerShell window and try:

```
psql --version  
pg_isready -h localhost -p 5432
```

If psql is not recognized, add this folder to your Windows PATH, then open a new PowerShell window:

```
C:\Program Files\PostgreSQL\18\bin
```

Create Local Database And App User

Log in as the postgres superuser:

```
psql -U postgres -h localhost -p 5432
```

Run the following SQL:

```
CREATE DATABASE immiauto;  
  
CREATE USER immiauto_user  
    WITH PASSWORD 'ChangeThisStrongPassword123!';  
  
GRANT ALL PRIVILEGES ON DATABASE immiauto TO immiauto_user;  
  
\c immiauto  
  
GRANT ALL ON SCHEMA public TO immiauto_user;  
ALTER DEFAULT PRIVILEGES IN SCHEMA public  
    GRANT ALL ON TABLES TO immiauto_user;  
ALTER DEFAULT PRIVILEGES IN SCHEMA public  
    GRANT ALL ON SEQUENCES TO immiauto_user;
```

Test the app user:

```
psql -U immiauto_user -h localhost -p 5432 -d immiauto
```

Spring Boot Project Configuration

This project already includes the PostgreSQL JDBC driver in backend/pom.xml. The application.properties file currently uses H2 for development and also contains a commented PostgreSQL example.

The existing dependency should look like this:

```
<dependency>

    <groupId>org.postgresql</groupId>

    <artifactId>postgresql</artifactId>

    <scope>runtime</scope>

</dependency>
```

Use these application.properties database settings:

```
spring.datasource.url=jdbc:postgresql://localhost:5432/immiauto

spring.datasource.driver-class-name=org.postgresql.Driver

spring.datasource.username=immiauto_user

spring.datasource.password=ChangeThisStrongPassword123!


spring.jpa.hibernate.ddl-auto=update

spring.jpa.show-sql=true

spring.jpa.properties.hibernate.format_sql=true

spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.PostgreSQLDialect


spring.h2.console.enabled=false
```

Run The Backend

```
cd C:\Users\mh200\Projects\Immigration-Consultation\backend

.\mvnw spring-boot:run
```

Connection Summary

```
Host: localhost

Port: 5432

Database: immiauto

Username: immiauto_user

Password: ChangeThisStrongPassword123!

JDBC URL: jdbc:postgresql://localhost:5432/immiauto
```

Use the dedicated `immiauto_user` for the application. Keep the postgres superuser only for administration.

Useful Local Commands

```
# Check server readiness

pg_isready -h localhost -p 5432


# Connect as app user

psql -U immiauto_user -h localhost -p 5432 -d immiauto


# List databases inside psql

\l


# List tables inside psql

\dt


# Quit psql

\q
```

Official References

PostgreSQL Windows installers: <https://www.postgresql.org/download/windows/>

PostgreSQL versioning policy: <https://www.postgresql.org/support/versioning/>

PostgreSQL license: <https://www.postgresql.org/about/licence/>

EDB PostgreSQL downloads: <https://www.enterprisedb.com/downloads/postgres-postgresql-downloads>